

# Weighing Isotopes: Calculating Average Atomic Mass

PhET Isotopes and Atomic Mass · PhET

Years 9-10

~30 minutes

Science · Chemistry

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/isotopes-and-atomic-mass>

Curriculum links: AC9S10U06 · AC9S9U06 · atomic structure · periodic table

## Learning intentions

- I can describe what makes isotopes of an element different from each other.
- I can calculate a weighted average atomic mass from isotope abundance data.
- I can explain how atomic mass values on the periodic table relate to naturally occurring isotope mixtures.

## Getting started

Open PhET Isotopes and Atomic Mass and go to the 'Mixtures' screen. Choose an element from the periodic table on the right, then use the neutron slider or bucket to build a sample containing a mix of isotopes on the scale. Keep the abundance readout panel visible as you work.

## Tasks

**1. Select an element with at least two stable isotopes. Add several atoms of two different isotopes to the scale. Record the mass number of each isotope and how many of each you added.**

---

---

**2. Using the counts from Task 1, calculate the percentage abundance of each isotope in your sample.**

Working:

---

---

---

**3. Calculate the weighted average atomic mass of your sample by hand: multiply each isotope's mass by its percentage abundance (as a decimal) and add the results together. Show your working.**

Working:

---

---

---

**4. Check the simulation's average atomic mass readout against your hand calculation. How close are the two values? Suggest one reason for any small difference.**

---

---

**5. Reset the sample and build a new mixture using three different isotopes of the same element. Record your data in the table below.**

| Isotope (mass number) | Number of atoms added | % abundance |
|-----------------------|-----------------------|-------------|
|                       |                       |             |
|                       |                       |             |
|                       |                       |             |

**6. Using your table from Task 5, calculate the new weighted average atomic mass by hand, then compare it to the simulation's readout.**

Working:

---

---

---

**7. Explain, in terms of protons, neutrons and electrons, why two isotopes of the same element have different masses but the same chemical properties.**

---

---

---

---

**8. Recall the atomic mass value shown on a standard periodic table for your chosen element. Explain why this value is rarely a whole number.**

---

---

### ***Extension (optional)***

Chlorine's periodic table mass is about 35.45, even though no single chlorine atom has that mass. Use the isotope mixture data in the simulation (or your own reasoning about abundances) to explain how a mixture of whole-number isotope masses can average out to a non-whole-number value.

---

---

---